

Computational Paleography and the Voynich Manuscript: A Technical Evaluation of the King-Andrisani Tironian Hypothesis and Contemporary Transcription Models

The Voynich Manuscript (Beinecke MS 408) represents perhaps the most significant challenge to modern cryptanalysis and medieval paleography. Carbon-dated to the period between 1404 and 1438, the codex contains approximately 240 pages of text in an unknown script, accompanied by enigmatic botanical, astronomical, and biological illustrations.¹ The absence of a bilingual "Rosetta Stone" has necessitated the development of sophisticated digital transcription systems and computational models to analyze the text's underlying structure. The prevailing academic consensus has long oscillated between the manuscript being a complex cipher, a lost natural language, or an elaborate medieval hoax.³ In 2021, a significant hypothesis proposed by Tim King and Alessandra Andrisani suggested that the manuscript is a medical manual for women's health written in a Vulgar Latin dialect, using a modified subset of Tironian Notes.⁵ This report provides an exhaustive technical analysis of this hypothesis within the context of current transcription standards, morphological research, and competitive computational frameworks.

1. The EVA Transcription System and Its Technical Limitations

Digital analysis of the Voynich Manuscript is predicated on the conversion of handwritten glyphs into a machine-readable format. This process, termed transliteration, is complicated by the fact that the phonetic and semantic values of the characters are unknown.⁶ The European Voynich Alphabet (EVA), introduced by Gabriel Landini and René Zandbergen in 1998, serves as the current industry standard.⁶

1.1 The Architecture of EVA

EVA was designed with the objective of providing a human-readable transcription that maps Voynich glyphs to Latin characters based on visual similarity.⁶ For instance, the glyph resembling a digit '8' is mapped to d, and the character resembling a digit '9' is mapped to y.⁸ This design allows researchers to discuss the text using a shared alphabet, though it introduces specific biases into the data. EVA assumes that certain complex shapes are composed of simpler, independent characters, which may or may not be true in the original scribe's

conceptualization.⁹

One of the primary structural challenges in EVA is the treatment of minims—the vertical strokes that form characters such as i, n, and m. In EVA, these are often decomposed into individual units (e.g., iin for a three-stroke character), whereas earlier systems like the Currier alphabet frequently treated these sequences as single units.⁶ This decomposition significantly impacts statistical entropy calculations, as the frequency of single-minim characters increases disproportionately compared to their likely semantic value.¹²

1.2 Historical Transcription Efforts and Accuracy

Before EVA, several other systems were developed, each offering a different perspective on the manuscript's graphetics. The Friedman First Study Group (FSG) in the 1940s utilized capital letters and numbers, emphasizing the distinction between "pedestalled" and "standard" gallows.⁶ Prescott Currier's transcription in the 1970s was the first to identify two distinct statistical "languages" (A and B) within the manuscript, which he suggested were written by different scribes.⁸

Transcription System	Developer	Key Characteristic	Limitations
FSG	Friedman Group	Used digits and capitals; identified "pedestals" as separate units (Z). ⁶	Limited coverage of rare glyphs; outdated format.
Currier	Prescott Currier	Differentiated sections by statistical profiles (A vs. B). ⁸	Context-dependent mappings; ambiguous space representation.
EVA	Landini & Zandbergen	"Look-alike" mapping; highly readable; extensible. ⁶	Decomposes ligatures into minims; potential over-segmentation. ¹⁰
v101	Glen Claston	Highly granular; captures thousands of unique variants. ⁹	Complexity makes it difficult for standard NLP tools; low adoption.

The transition between these systems is often facilitated by tools such as Jacques Guy's "bitrans," which allows for the conversion of transliteration files between different alphabets using complex rule sets.⁹ However, even with these tools, the accuracy of transcriptions remains unknown, as the underlying "alphabet" of the manuscript has never been verified.⁶

1.3 Transcription Challenges: The "Weirdo" Codes

The Voynich script contains a subset of characters that appear with extreme rarity, often referred to as "weirdo codes".⁸ These glyphs, such as the "picnic table" (\$ \chi \$) or elaborate variants of the gallows characters, often occur only once or are concentrated on specific pages.⁸ The debate persists as to whether these are unique semantic units, scribal errors, or decorative flourishes. In EVA-based research, these are often handled using "weirdo codes" (e.g., &NNN;), which prevents them from being collapsed into common characters during statistical analysis.¹³

Furthermore, word spacing in the manuscript is notoriously ambiguous. While the text appears to be divided into words, the gaps between characters are not always consistent.⁶ Some transcriptions introduce a comma to designate an "uncertain space," which shifts the problem of word identification from a binary decision to a probabilistic one.¹⁰ This is particularly relevant for computational approaches, as incorrect word segmentation can skew Zipf's Law and entropy measurements.³

2. Glyph Analysis: State of the Art in Voynich Paleography

Contemporary analysis of the Voynich glyph system has moved beyond simple character identification toward a holistic understanding of the script's anatomy and scribal execution. Paleographers have identified several distinct categories of characters that define the script's visual and structural identity.

2.1 The Gallows and Pedestals

The "gallows" are a set of four prominent characters—transcribed in EVA as f, p, k, t—that ascend significantly above the height of other characters.⁶ These are considered the structural backbone of the script. They frequently appear in word-initial positions and are subject to specific positional rules.⁸

- **Standard Gallows:** The basic forms of f, p, k, t.
- **Pedestalled Gallows:** These occur when a gallows character is placed atop a "bench" or a "platform" (typically transcribed as the ch sequence in EVA, forming units like cth or ckh).⁸
- **Intruding Gallows:** Some transcriptions, particularly the FSG, noted cases where gallows appeared to "intrude" into other words or were merged with existing sequences.⁶

An interesting statistical anomaly exists regarding the gallows: while t and k are followed by the

glyph e approximately 50% of the time, the more elaborate versions p and f are virtually never followed by e.¹¹ This suggests that p and f are not merely decorative variants of t and k, but distinct units with different grammatical or phonetic roles.¹¹

2.2 Bench Characters and Loops

The "bench" characters, primarily ch and sh, often function as prefixes or components of the pedestalled gallows.⁸ These characters are composed of a "c-curve" combined with a vertical or looped element. Paleographic study indicates that the script utilizes two main series of shapes: those starting with a "c-curve" (including a, o, d, y) and those starting with an "i-minim" (including i, n, m, l, r).⁸

The role of "loops" is another point of intense study. Many characters in the manuscript feature a terminal loop, such as EVA y, m, and g. In traditional Latin paleography, such loops often represent abbreviations for common suffixes like *-us*, *-is*, or *-em*.⁸ This has led many researchers to conclude that the Voynich script is heavily abbreviated, potentially making it a form of "shorthand" rather than a standard alphabet.¹

2.3 The Five Scribal Hands

Research by Prescott Currier and later refinements by other scholars have identified at least five distinct scribal "hands" in the manuscript, each associated with specific sections and linguistic properties.⁸

Hand	Primary Content Area	Language Profile	Characteristics
Hand 1	Herbal-A and Pharmaceutical sections.	Currier Language A.	More restricted vocabulary; simpler word structures. ⁸
Hand 2	Herbal-B and Biological sections.	Currier Language B.	Higher entropy; more frequent "unattached finals". ⁸
Hand 3	Starred paragraphs (Section 5).	Transitions between A/B.	Dense text blocks with stars in the margin. ⁸
Hand 4	Astronomical and Zodiac sections.	Unique "labelese" vocabulary.	Often written in circles or radii; integrates text with

			diagrams. ⁸
Hand 5	Subset of Herbal-B and Quire 8.	Language B.	Smaller sample size but consistent with Hand 2. ⁸

The existence of multiple hands suggests that the manuscript was a collaborative effort, possibly produced in a scriptorium or a dedicated scholarly environment. This collaboration implies that the writing system was a standardized, learnable code or language used by a group of people, rather than a purely idiosyncratic personal shorthand.¹⁷

3. The King-Andrisani Hypothesis (2021)

In 2021, Tim King and Alessandra Andrisani proposed a detailed decryption framework that identifies the manuscript as a manual for medical practices, specifically focusing on women’s health.⁵ Their theory rests on two primary pillars: the underlying language is a Vulgar Latin dialect from Northern Italy, and the script is a modified adaptation of Tironian Notes.⁵

3.1 Linguistic Foundation: Vulgar Latin and the Veneto Dialect

King and Andrisani argue that the text reflects the linguistic shifts characteristic of the transition from Latin to the Romance languages, specifically influenced by the dialects of the Veneto region (Verona, Padova, and Vicenza).⁵ They identify several phonetic shifts in the transliterated text that mirror documented historical changes:

- The shift of Latin *illo* (that) to *elo*.⁵
- The shift of Latin *iis* (those) to *ies*.⁵
- The use of phonetic spellings for "x" as /ks/ (e.g., *ec.çi.o.e* for *excioe*), as the script lacks a dedicated glyph for the letter "x".⁵

The content is interpreted as a "manual for women's health," a reading supported by the biological section's illustrations, which King and Andrisani view as depicting medicinal bathing and pharmacological processes related to gynecological care.⁵ They link the manuscript to the intellectual tradition of 15th-century Italy, where humanist hands were used for classical and scientific texts.⁵

3.2 Phonetic Mapping and Sound Values

The King-Andrisani model proposes twenty-two distinct characters, emphasizing that several are ligatures or confluations.⁵ They assign phonetic values to EVA characters that diverge from the standard "look-alike" mappings, aiming for a functionally coherent reading of the text.

EVA Glyph	King-Andrisani Phonetic	Proposed Meaning/Role
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	Value	
p	<i>pe</i>	Preposition <i>per</i> or the syllable /per/. ⁵
q	<i>co</i>	Conjunction <i>con</i> or the syllable /con/. ⁵
y	<i>un / in / nu</i>	Functions as sound value, punctuation, or truncation. ⁵
8	<i>et</i>	Corresponds to the Tironian note for "and". ¹⁸
9	<i>us</i>	Terminal abbreviation for the Latin suffix <i>-us</i> . ⁸
d	<i>de</i>	Functions as a sound value and a marker for truncation. ⁵

Using this mapping, the authors provide translations of several pages. For instance, the string i.a.d.u.ure on Folio 1v is read as *adure* (Latin for "to burn" or "cauterize"), and ion on Folio 9v is interpreted as *viola* (the flower depicted on the page).⁵ The hypothesis claims that these mappings reveal a text that identifies medicinal plants and their therapeutic uses, often citing historical precedents such as Ibn Sina's *Materia Medica* (ca. AD 1000).⁵

3.3 Critique of the King-Andrisani Hypothesis

While the King-Andrisani thesis provides a cohesive narrative and matches the carbon-dating context, it has been met with significant skepticism from the Voynich research community.¹⁶

Critics argue that the model relies on "polyvalence"—assigning multiple sound values to a single glyph—which introduces excessive subjectivity. If the character y can represent /n/, /s/, /nu/, or /un/ at the researcher's discretion, it becomes possible to force almost any Latin-like word from a given string.¹⁶ This subjectivity is a common pitfall in Voynich research, most notably seen in the controversial work of Gerard Cheshire.¹⁹

Furthermore, the phonetic frequency distribution of the proposed mappings does not align with natural Latin. For instance, y is one of the most common characters in the manuscript. If it represents sounds like /in/ or /un/, the resulting Latin text would have a disproportionately high frequency of these endings compared to the common Latin suffixes *-us*, *-um*, *-em*.¹⁶ The absence of a dedicated "x" is also problematic, as "x" is highly frequent in medieval Latin medical terminology, and phonetic spelling as /ks/ would significantly alter the expected word

length distributions.⁵

4. The Tironian Notes Connection

The claim that the Voynich script is derived from Tironian Notes is a core component of the King-Andrisani hypothesis. Tironian Notes (*notae Tironianae*) was an ancient Roman shorthand system invented by Marcus Tullius Tiro in the 1st century BC.²⁰

4.1 History and Mechanics of Tironian Shorthand

The system was originally designed to record the speeches of Cicero and consisted of roughly 4,000 signs.²⁰ During the medieval period, particularly under the Carolingian dynasty (750–900 AD), the system was expanded to over 13,000 signs and used extensively in monasteries for copying religious and everyday documents.²⁰ Tironian notes are often "composites" or ligatures, where a simpler sign is modified with various marks to form a complex idea or a specific syllable.²⁰

- **The Tironian '7':** The sign for *et* (and) is perhaps the most famous and longest-lived Tironian note. It resembled a digit '7' (or sometimes '8') and remained in use in normal Latin long-hand and in Irish and Scottish Gaelic for centuries.²⁰
- **The '-us' Sign:** A sign resembling a '9' was commonly used to represent the suffix *-us*.⁸

4.2 Systematic Comparison with Voynich Glyphs

The Voynich script shares certain visual elements with Tironian and other medieval shorthand systems. Many researchers have pointed out that the Voynich 8 and 9 are nearly identical to the Latin abbreviations for *et* and *-us*.⁸ However, the connection beyond these common symbols is debated.

A detailed critique from the *Voynich Ninja* forum suggests that many of the characters King and Andrisani identify as Tironian are actually standard Latin or Greek scribal abbreviations.¹⁶ For example, the upper-right loop they identify as Tironian "e" is more likely a common Greek convention for /is/ or /o/.¹⁶ Tironian shorthand was generally more abstract and less "word-like" than the Voynich script, which appears to follow a very specific "slot grammar".¹³

Schmitz's *Commentarii Notarum Tironianarum* (1893) remains the primary reference for these signs.²² A systematic comparison shows that while the Voynich script may have been *inspired* by shorthand systems or used a few of their more common symbols, it does not strictly follow the Tironian conventions.⁸ Instead, it appears to be a unique creation that adapts various medieval elements—Latin abbreviations, Arabic numerals, and alchemical symbols—into a new, cohesive system.⁸

5. Morphological Structure: The Stolfi Paradigm

To validate any decryption, one must account for the rigorous morphological structure of the

Voynich "words." This structure is best described by the Stolfi Paradigm, which models each word as a sequence of three "slots": Prefix, Stem, and Suffix.¹³

5.1 The Prefix-Stem-Suffix Model

Statistical analysis shows that the distribution of glyphs within a word is not random. Characters like qo nearly always appear at the beginning of a word, while characters like m or g are almost exclusively found at the end.¹³

Component	EVA Glyphs	Estimated Function
Prefix	qo, sh, ch	Operators, target markers, or grammatical particles. ¹⁵
Stem	o, a, ee	Vowel cores; define the "center" of the word. ¹⁵
Suffix	edy, ain, al, ol	Grammatical markers for processing stage, result, or category. ¹⁵

This rigid structure is extremely unusual for a natural language. In Latin, for instance, prefixes and suffixes are common, but the internal "stems" allow for a wide variety of consonant and vowel combinations. The Voynich script, by contrast, behaves more like a "procedural" code or a highly regular artificial language.⁴

5.2 The sh→ch Transformation and the "Triangular Cycle"

In the pharmaceutical and recipes sections, researchers have identified what appear to be functional grammar rules for material processing.¹⁵

1. **sh-words vs. ch-words:** Statistical profiles suggest that sh- prefixes denote "primary materials" or raw ingredients, while ch- prefixes denote "processed products".¹⁵
2. **The Processing Cycle:** A "triangular cycle" of suffixes has been identified—edy (basic processing), eedy (refinement), and ain (output/result).¹⁵
3. **Operator Patterns:** The text follows an alternating pattern of "operator → modifier → reference" (e.g., ok names/transforms, ot refers/observes).¹⁵

The common sequence daiin (or aiin) is viewed as a "universal connector" or a marker of iterative operations.¹⁵ This suggests that the manuscript is not merely a collection of names and descriptions, but a series of operational procedures—essentially "recipes" for creating pharmaceutical compounds or performing medical treatments.¹⁵

6. Competing Hypotheses and Computational Evaluations

The King-Andrisani hypothesis is one of several major frameworks attempting to solve the manuscript. Computational evaluations have provided quantitative metrics to compare these theories.

6.1 The Naibbe Cipher (Greshko 2025)

Michael Greshko's 2025 proposal presents a "verbose homophonic substitution cipher" called the Naibbe cipher.²³ This model suggests the manuscript is a ciphertext of Latin or Italian, encrypted using a method plausible for the 15th century (likely involving playing cards as a randomizing element for selecting substitution tables).²⁴

The Naibbe cipher is significant because it is the first model to successfully replicate many of the manuscript's "anomalous" statistics:

- **Zipf's Law:** It reproduces the word frequency distribution ($R^2 \geq 0.94$).¹⁵
- **Heaps' Law:** It matches the vocabulary growth rate and the high number of *hapax legomena* (words that appear only once).²⁴
- **Word Families:** It explains why Voynich words are often clustered into "families" that differ by only a single character.²⁴

Unlike King and Andrisani's "shorthand" theory, which assumes a somewhat direct linguistic relationship, the Naibbe cipher posits a "verbose" relationship where a single plaintext letter maps to a multi-glyph Voynich "word" or string.²⁴ This accounts for the low conditional entropy of the manuscript, as the "slot grammar" becomes an artifact of the encryption process rather than a property of the source language.²⁴

6.2 The Hebrew Consonantal Cipher

Another major hypothesis proposes that the manuscript encodes Hebrew text.¹⁸ A 2021 study evaluated a mapping of 19 EVA characters to 19 Hebrew consonants.²⁶

Metric	Voynich (EVA)	Natural Hebrew	Decoded Hebrew Cipher
Entropy (h_2)	2.12 bits	3.72 bits	2.44 bits

Index of Coincidence (IC)	0.077	0.077	0.077
Lexical Match (z-score)	N/A	N/A	3.6 to 4.4

While the statistical signal for Hebrew is strong (with z-scores suggesting the mapping captures genuine structural correspondences), the decoded text does not form coherent, readable sentences.²⁶ This suggests that if the manuscript is a Hebrew cipher, it is either more complex than a monoalphabetic substitution or is a "near-miss" where the frequency profile matches Hebrew, but the sequential structure is still obscured by another layer of encoding.²⁶

6.3 The Nahuatl and Aztec Theory

The hypothesis that the manuscript is written in Nahuatl (Aztec) relies heavily on the identification of New World flora, such as the sunflower (*Helianthus annuus*).¹⁷ Linguistically, Nahuatl is an agglutinative language where morphemes are combined into complex words, which some argue fits the "compound word" structure of Voynichese.¹⁷ However, this theory faces a major chronological hurdle: the 1404–1438 carbon date precedes European contact with the Aztecs by nearly a century.² While some proponents suggest the vellum was "old stock," the art style remains distinctly European.²

7. Recent Computational Approaches (2020–2025)

The last five years have seen an influx of high-level NLP and machine learning techniques applied to MS 408.

7.1 Hidden Markov Models and Phonology

Unsupervised learning algorithms, such as Hidden Markov Models (HMM), have been used to analyze the "word grammar" of the manuscript.²⁸ A 2-state HMM found that the last character of every word is generated by a specific state, while all other characters are generated by another, creating a grammar described as $a * b$.²⁸ This pattern is observed in abjads (scripts without vowels like Arabic or vowelless English), which supports the idea that the Voynich script might omit vowels—a premise used by both the Hebrew and the Latin-shorthand theories.¹⁴

7.2 Probabilistic Latin Models (Shir-Man 2025)

A 2025 computational study by Shir-Man utilized Expectation-Maximization (EM) iterations to test a "probabilistic key" for Latin.³⁰ This model used a 300-dimensional semantic embedding of

herbal and alchemical Latin to evaluate potential decryptions.

- **Morphological Parse Success:** 78.6% for a subset of medieval Latin.³⁰
- **Semantic Coherence:** 0.62 (compared to a random baseline of 0.19).³⁰
- **Topic Coherence:** The resulting text showed domain-specific consistency with 15th-century medical herbals.³⁰

This model suggests that the text could be decrypted if one assumes a "probabilistic" mapping where common Voynich suffixes like *y* correspond to Latin genitive or plural endings like *-is* or *-um*.³⁰ This supports the King-Andrisani idea of a Latin base, but suggests a more complex, abbreviation-heavy structure than a simple shorthand.

7.3 Entropy and Burstiness Analysis

Landini and others have applied Zipf's Law and "Poissonian burstiness" to characterize the text's information-encoding system.³ Natural languages typically exhibit "burstiness," where certain words appear in clusters related to specific topics.¹⁵ The Voynich Manuscript shows a burstiness value of $B \approx 0$, indicating an extremely regular, almost mechanical distribution of words.¹⁵

This regularity is a "red flag" for natural language. It suggests that the text is either an extremely repetitive religious or technical document (like a formulary or a list of procedures) or that it was generated by an algorithm that enforces a specific "slot grammar".¹³

8. Conclusions and Implications for the King-Andrisani Hypothesis

The technical validation of the King-Andrisani hypothesis requires it to account for both the paleographic history of Tironian Notes and the modern statistical properties of the "Voynichese" text.

8.1 Synthesis of Findings

1. **Contextual Match:** The 15th-century Northern Italian (Veneto) setting and the "medical manual" purpose align perfectly with carbon dating and iconographic evidence.²
2. **Shorthand Viability:** The manuscript's script shares common abbreviation markers with Tironian Notes and medieval Latin long-hand, supporting the "abbreviated Latin" theory.⁵
3. **Statistical Challenges:** The King-Andrisani model, like many "natural language" decryptions, struggles with the manuscript's anomalously low entropy and rigid slot grammar. Natural Vulgar Latin, even with abbreviations, rarely exhibits the $a * b$ HMM structure or the $B \approx 0$ burstiness found in MS 408.¹⁵
4. **The Cipher Alternative:** Models like the Naibbe cipher (2025) provide a compelling

alternative explanation for the "slot grammar," suggesting it is a byproduct of a homophonic substitution system rather than a property of the language itself.²⁴

8.2 Actionable Considerations for Decryption

To advance the computational decryption of the manuscript, a researcher must move beyond the "look-alike" EVA transcription and adopt a model that accounts for the "polyvalence" and "abbreviation" suggested by King and Andrisani, while maintaining the statistical rigor required by modern linguistics. The "sh→ch" material transformation patterns and the "edy→eedy→ain" processing cycles offer a functional "crib" that may be more valuable for decryption than character-by-character substitution.¹⁵

Ultimately, the manuscript appears to be a "composite knowledge document".¹⁵ Whether it is a direct transcription of a Tironian-style shorthand or a more complex verbose cipher, it represents a sophisticated 15th-century attempt to encode technical information—likely related to medicine and pharmacology—in a system that has successfully defied simple cryptanalysis for over six hundred years. Further validation of the King-Andrisani hypothesis will require a larger, team-based effort involving specialists in Vulgar Latin, medieval medicine, and computational linguistics to verify the semantic coherence of the proposed translations across the entire 240-page corpus.⁵

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